

# Arka Prava Sarkar

Ph.D.

✉ [sarkar.arkaprava200@gmail.com](mailto:sarkar.arkaprava200@gmail.com)  
DOB: 25th of July, 1994



## Work Experience

- August 2026–Present **Quantum R&D Consultant, QuLattice, ScInter Pty Ltd..**  
Development of adaptive noise modeling for QEC
- January 2026–Present **Post Doctoral Research Associate, Condensed Matter and Materials Physics, Satyendra Nath Bose National Centre for Basic Sciences, Kolkata.**  
Current project: Optimizing hybrid perovskites Solar Cells  
**Advisor : Sr. Prof. Priya Mahadevan**
- Google scholar:** <https://scholar.google.com/citations?user=kfJzDJMAAAAJ&hl=en>

## Education

- 2019–2026 **Ph.D., Computational Materials Science, Indian Institute of Technology, Kharagpur.**  
Thesis Title : Computational Insights into Ionic Liquid Adsorption on Metallic Surfaces: Dynamics, Orientation and the Influence of Surface Heterogeneity.  
**Advisor : Dr. Sandeep Kumar Reddy**
- 2016–2018 **M.Sc Physics, Indian Institute of Technology (ISM), Dhanbad, OGPA–9.06.**  
M.Sc Thesis on Computational Soft Condensed Matter Physics.  
Elective Courses : Astrophysics and Cosmology, High Energy Physics and Computational Physics.
- 2012–2016 **B.Sc Honours, University of Calcutta, Kolkata, Percentage–60.00%.**  
Physics Honours, Mathematics and Computer Science
- 2011–2012 **XIIth Board, Kendriya Vidyalaya Ordnance Factory, Dum Dum, Percentage – 87.00%.**  
Mathematics, Physics, Chemistry and Computer Science
- 2009–2010 **Xth Board, Kendriya Vidyalaya Ordnance Factory, Dum Dum, Percentage – 85.50%.**  
Secondary School Examination

## Research Interest

- Quantum error correction and noise modelling
- Adsorption dynamics and stability of ionic liquid-metal surface based complex systems using classical and quantum simulations
- Optimization of hybrid perovskites solar cells for various applications using first principles simulations
- Quantum computation based machine learning techniques for biological markers and disease predictions

## Core Scientific Expertise

- Density functional theory
- Replica exchange molecular dynamics simulation
- Quantum Computing

- Two dimensional materials and metal surfaces
- Solid-liquid interfaces
- Computational materials science
- Quantum mechanics
- Statistical mechanics

## Programming and Scientific Computing

- Scientific coding for data analysis
- Programming languages: C++, Python, Bash scripting (Intermediate level)
- Libraries: NumPy, Matplotlib, pandas, SciPy, RDKit, Seaborn, Qiskit, etc.

## Computational Tools and Softwares

- DFT: CP2K, Gaussian16, VASP, Quantum Espresso, ASE
- Molecular Dynamics: GROMACS, LAMMPS
- Data Analysis and Visualization: VMD, GaussView, ASE, OVITO

## HPC Skill

- Operating System: Linux and Unix

## List of Publications

- **Sarkar, A.P.**, Investigating the Role of Anions in the Adsorption of Pyrrolidinium Based Ionic Liquids on Pt(111) Surface Using Density Functional Theory. *Int. J. Quant. Chem.*, **2024**, 124, 20, e27497. <https://doi.org/10.1002/qua.27497>
- **Sarkar, A.P.**, Reddy, S.K., Adsorption of Imidazolium-Based Ionic Liquid On Pt(111) Surface Studied Using Density Functional Theory. *Adv. Theo. Simul.*, **2025**, 8, 2, 2400458. <https://doi.org/10.1002/adts.202400458>
- **Sarkar, A.P.**, Sahu, R., Giri, S., Anakuthil, A., Reddy, S.K., Exploring the conformational space of sulfonyl-based ionic liquid on platinum based mono and bimetallic surfaces. *Langmuir*, **2025**, 41, 5, 3187-3198. <https://doi.org/10.1021/acs.langmuir.4c03883>
- **Sarkar, A.P.**, Saleh, M., Reddy, S.K., Influence of metallic surface heterogeneity on adsorption dynamics and orientation of sulfonyl-functionalized ionic liquids. *Langmuir*, **2026**, 42, 6, 4839-4849. <https://doi.org/10.1021/acs.langmuir.5c05813>
- Gupta, M.G., **Sarkar, A.P.**, A Curvature Guided Composite Kernel Framework for Differential Gene Selection in Cancer Transcriptomics. (BioRxiv preprint: <https://doi.org/10.64898/2026.07.23.740443>)
- **Sarkar, A.P.**, Surface-Termination, Strain Engineering and Vacancy Effects of  $\text{Ti}_3\text{C}_2\text{T}_x$  MXenes for Broadband Mid-Infrared Reflective Coatings Using Density Functional Theory. (Under Peer Review)

## Teaching Experience

- 2024 **Research project guidance**, *Indian Institute of Technology*, Kharagpur, Work : guiding undergraduate summer intern for research work.  
Optimizing structure-property relations of 2D materials using density functional theory.
- 2020-2025 **Teaching Assistantship**, *Indian Institute of Technology*, Kharagpur, Course : High Performance Computing and it's Applications to Complex Physical Systems.  
Conducting hands on session for running Molecular Dynamics simulation of different complex physico-chemical systems and to compute different thermodynamic properties. Parallelized versions of the codes such as LAMMPS and GROMACS are executed on HPC clusters and supercomputer. Theories related to Molecular Dynamics and Monte Carlo simulations along with elementary force field parameters are also discussed as a part of the teaching plan. Additionally, the teaching plan also involves discussions related to parallel programming based on FORTRAN, CPU-GPU architecture, etc.

2023-2024 **Teaching Assistantship**, *Indian Institute of Technology*, Kharagpur, Course : Quantum methods in Molecular Simulations.

Conducting theoretical lectures and hands on session for Density Functional Theory (DFT) of different complex chemical and materials systems and to compute different ground state properties. Parallelized versions of the DFT codes such as Gaussian16 and CP2K are executed on HPC clusters and supercomputer.

2017–2019 **Online Mentoring**, *IL & FS Education*, Kolkata, Subject: Science.

Helping students of secondary level (Class 8, 9 and 10) of CBSE Board via Geneo App for Physics subject; creating and editing MCQ and subjective questions of CBSE pattern; AI text curation, MCQ curation; Creating/editing sample papers and solving questions according to the latest pattern CBSE class 10 board exam papers of science subject.

## Conferences and Workshops

- Participated and presented poster at the PCCP SPHERE 2025 international conference, organised by IISc, Bengaluru.  
Duration: 16th to 18th October 2025
- Participated and presented poster at the national conference on Current Trends in Theoretical Chemistry, organised by BARC, Mumbai.  
Duration: September 2024
- Participated and presented poster at the CECAM School, titled "**Long time multi-scale simulation of activated events: from theory to practice**" at SISSA, Italy.  
Duration: June 23-28, 2024
- Participated and presented poster at the MD@60 international conference jointly organised by JNCASR and CECAM.  
Duration: 25th to 29th February, 2024
- **Co Instructor** on High performance Computing workshop on Bio-informatics  
Jointly organised by National Supercomputing Mission, CDAC and CCDS, IIT Kharagpur.  
Duration: 10th and 11th February, 2024
- International workshop and conference on Evolution of Electronic Structure Theory and Experimental Realization (EESTER).  
Jointly organised by SRM Institute of Technology, Kattankulathur and IIT Madras, Chennai  
Duration: 4th to 12th January 2023
- 20th International Workshop on Computational Physics and Materials Science: Total Energy and Force Methods (Online Mode).  
Organized by International Centre for Theoretical Physics, Italy  
Duration : February 2021
- Online Workshop On Accelerated Data Science.  
Organised by Indian Institute of Technology, Kharagpur  
Duration : February 2021
- User Workshop on Supercomputing  
Organised by Centre for Development of Advanced Computing, India  
Duration : November 2020
- National Conference on Advancements in Spectroscopic Techniques and Materials.  
Organized by Department of Applied Physics, IIT (ISM) Dhanbad  
Duration: March 2018

## Achievements

GATE 2019 **Subject : Physics**, GATE Score : 467.

IELTS 2018 **Academic**, 2018, Overall: 8.0, CEFR Level: C1.

## Languages

Bengali Mother tongue

English Intermediate

*Conversationally good*

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## References

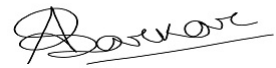
- Sr. Prof. Priya Mahadevan  
Email: priya@bose.res.in
- Dr. Sandeep Kumar Reddy  
Email : skreddy@iitkgp.ac.in
- Dr. Divya Nayar  
Email : Divya.Nayar@mse.iitd.ac.in
- Prof. Anakuthil Anoop  
Email: anoop.a@duk.ac.in
- Dr. Amit Kundu  
Email : amit.iop@yahoo.com

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## Declaration

I hereby declare that the above mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above mentioned particular.

**Place:** Kolkata



*Arka Prava Sarkar*